SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT SIH26181 · MASTER ENGINEERING MANUAL

VALOR: Vital and Atmospheric Logic for Offline Rescue

The Restructured Engineer's Guide & Jury Defense Manual — Dual-Brain FPGA + ESP32-S3 Edge SoC, Zero-Cloud INT8 TinyML & Multi-Hazard Clinical Triage

Attribute	Specification & Scope
System Class	Heterogeneous edge wearable: Renesas ForgeFPGA SLG47910 (Verilog RTL @ 50 MHz) + ESP32-S3 (Dual-Core FreeRTOS C11) + Zero-Cloud INT8 TinyML (619 bytes) + Deterministic mNEWS2 Safety Envelope.
Golden Reference vs. Deployable Target	**Golden Reference:** AMD Xilinx Zynq-7000 (`xc7z020clg400-1` , AXI4-Lite, 185 LUTs, WNS +5.603 ns). **Deployable Demo:** Vicharak ShrikeFi (ESP32-S3 + ForgeFPGA SLG47910, 4-wire SPI Mode 0, 515/1120 LUT5s, 105/140 CLBs, 0 DSP, 0 BRAM, WNS +7.316 ns, F_{max} 78.85 MHz).
Core Engineering Thesis	Commercial smartwatches fail when disaster destroys cellular backhaul, and RTOS scheduling jitter (5–10 ms) corrupts beat-to-beat heart-rate variability (RMSSD). VALOR timestamps systolic peaks in dedicated FPGA hardware at 20 ns resolution and fuses vitals with ambient hazards 100% offline.

MANDATORY REPORTING RULE FOR EVERY NUMBER IN THIS MANUAL

Always quote the MIMIC-III v1.4 benchmark (16,387 ICU records across 98 patients) as a paired result: **58.18% sensitivity / 98.19% specificity** alongside the **94.11% overall accuracy**. Because only 10.2% of records are positive alerts, a naive "always normal" classifier already scores 89.8%. Furthermore, never present the peripheral fingertip skin-temperature estimate as a core body temperature; the mNEWS2 temperature score is deliberately held at 0.

0. How to Use This Restructured Manual & Reading Paths

This manual consolidates the entire VALOR hardware, firmware, machine learning, and clinical defense into five streamlined parts. You do not need to memorize pin numbers or hexadecimal memory maps. If you can explain **why the hardware is partitioned across two chips**, **how signals move through the 13-step pipeline**, **how each finite state machine operates**, and **where the honest engineering boundaries lie**, you can defend VALOR before any technical jury.

Preparation Scenario	Chapters to Read	Focus & Deliverable
First Complete Read (~45 min)	Chapters 1–4, 6–10	End-to-end architecture: disaster context → SPI link → 8-tap filter → Peak-Detector FSM → FreeRTOS dual-core split → INT8 TinyML → mNEWS2 rules.
Deep Hardware / RTL Jury Review	Chapters 3–7, 12, 14	Zynq vs. ForgeFPGA synthesis, SPI Mode 0 timing, 2-FF CDC synchronizers, Verilog shift-register rules, Moore FSM encoding, CLB vs. LUT constraints, WNS +7.316 ns.
Pre-Demo Stage Rehearsal (30 min)	Chapters 13, 14, 15 & App. A–B	7-minute 5-act demo script, stage failure recovery, honest limitations, unified reconciliations, and the 30-question Judge Q&A Playbook.

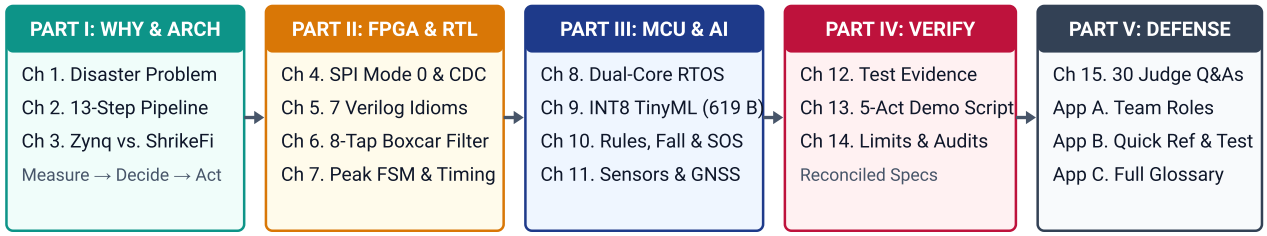


Figure 0.1 — Restructured roadmap of the VALOR manual. Redundant chapter-end Q&As and scattered editing notes have been unified into dedicated defense and reconciliation chapters.

PART I · WHY & WHAT

The Disaster Problem, System Architecture & Dual Hardware Targets

1. The Disaster Problem & Why Single-Sensor Alarms Fail

When a tropical cyclone, flash flood, industrial fire, or earthquake strikes, backhaul fiber tears, cellular base stations lose grid power, and cloud APIs become unreachable. A consumer wearable that depends on smartphone tethering or cloud inference becomes inert at the exact moment casualties mount. Simultaneously, extreme environmental hazards accelerate human physiological collapse.

Environmental Hazard	Physiological Mechanism of Injury	What VALOR Fuses On-Device
Heat & Humidity	Above ~40 °C with high relative humidity, sweat cannot evaporate; core temperature rises rapidly toward heat exhaustion and heat stroke.	BME280 temperature + humidity → NOAA Rothfusz Heat Index (clamped at 58 °C) fused with Heart Rate and RMSSD.

Environmental Hazard	Physiological Mechanism of Injury	What VALOR Fuses On-Device
Toxic Smoke (PM2.5)	Fine combustion particles (≤ 2.5 µm) cross the alveolar-capillary barrier, causing hypoxemia, airway inflammation, and autonomic depression.	PMS5003 laser scattering (EPA/AirGradient polynomial corrected) fused with SpO_2 desaturation and tachycardia.
Falling Barometric Pressure	A rapid synoptic pressure drop signals an approaching cyclone eyewall or severe squall line hours before peak wind impact.	BME280 pressure linear least-squares slope over a 180-minute rolling window (hPa/h) with a 0.5 hPa minimum drop noise floor.
Cold-Water Immersion	Floodwater immersion triggers cold shock, peripheral vasoconstriction, hypothermia, and late-stage bradycardia.	Cold-stress ambient exposure proxy fused with heart rate (deliberately capped at HIGH because no core-temperature probe is present).

The Alarm-Fatigue Trap of Single-Domain Devices

A conventional fitness band monitors only heart rate: **135 BPM** while jogging in a cool 24 °C park is healthy cardiovascular exertion, whereas the exact same **135 BPM** at 44 °C in heavy smoke with depressed RMSSD is impending heat stroke. Conversely, a stationary air-quality monitor measures city pollution but knows nothing about whether a rescuer is desaturating. If a device fires simple static threshold alarms during routine physical work, responders suffer **alarm fatigue** and turn it off. VALOR prevents alarm fatigue by evaluating vitals *in the context of* ambient thermal, particulate, and kinematic stress.

2. System Architecture & The 13-Step Signal Pipeline

VALOR operates as a closed, 100% offline **Measure** → **Decide** → **Act** loop partitioned across two silicon chips: anything requiring cycle-accurate, deterministic timing runs in dedicated FPGA hardware (50 MHz), while floating-point sensor fusion, neural inference, clinical rules, and local Wi-Fi hosting run on the dual-core ESP32-S3 microcontroller (up to 240 MHz).

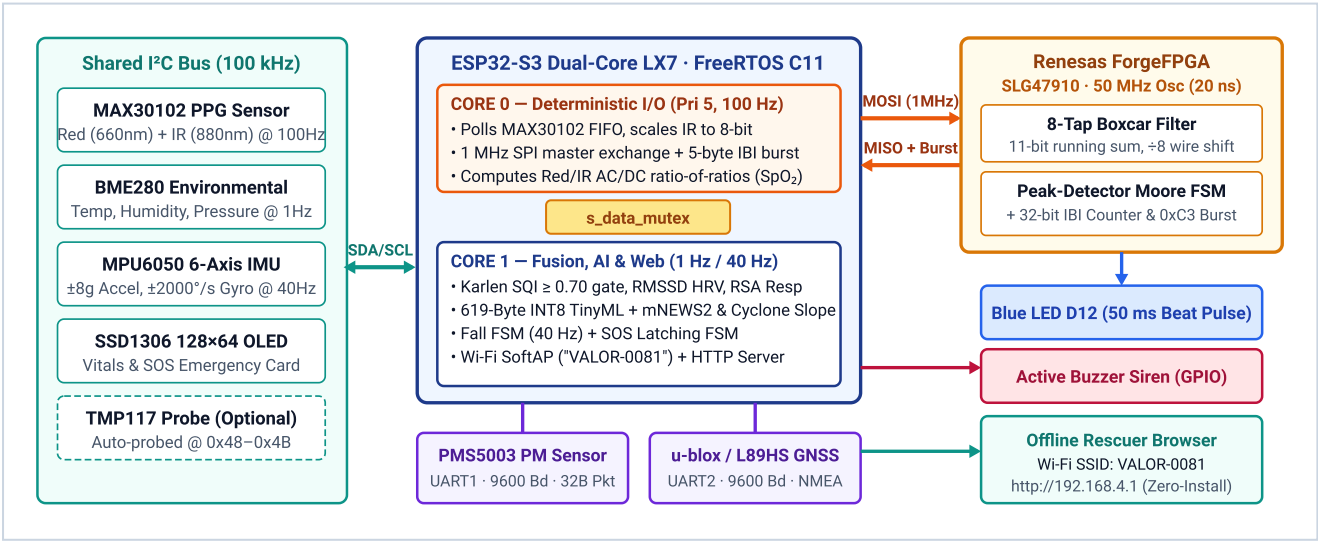


Figure 2.1 — Complete VALOR hardware architecture. Color coding: Teal = shared 100 kHz I²C sensor bus; Orange = 1 MHz full-duplex SPI link; Amber = 50 MHz FPGA fabric; Purple = asynchronous 9600-baud UART streams.

Tracing One Heartbeat from Photon to Local Alert (13 Steps)

When a judge asks how a single heartbeat travels through the wearable, trace these thirteen sequential stages across the I²C bus, ESP32 Core 0, the SPI bridge, the ForgeFPGA fabric, and ESP32 Core 1:

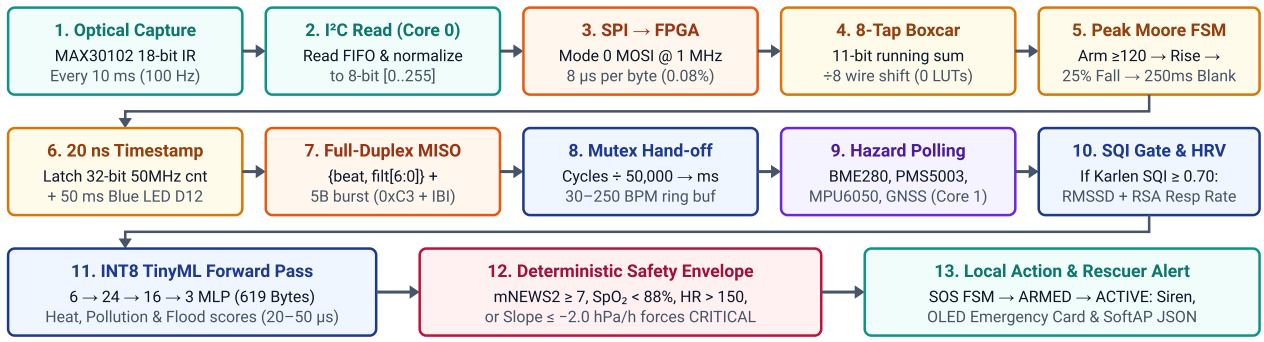


Figure 2.2 — The 13-step heartbeat and hazard pipeline from photodiode sample to local emergency alert.

#	Pipeline Stage	Hardware / Firmware Execution Detail
1	Optical capture	Every 10 ms (100 Hz), the MAX30102 captures an 18-bit ADC reading of reflected 880 nm infrared and 660 nm red light from the fingertip capillary bed.
2	I²C ingestion (Core 0)	`task_ppg_accelerator` (Priority 5) reads the FIFO over 100 kHz I²C, caches the red sample in software for SpO_2, and scales the IR sample to an 8-bit unsigned byte.
3	SPI to ForgeFPGA	Core 0 asserts `spi_ss_n` LOW and clocks the 8-bit IR sample out on `spi_mosi` at 1 MHz (Mode 0, 8 µs transfer duration).
4	Hardware filter	`moving_average_8tap.v` maintains an 11-bit running sum of the last 8 samples and outputs the upper 8 bits (`next_sum[10:3]`), dividing by 8 with zero logic gates.
5	Peak-Detector FSM	`ppg_peak_detector.v` (4-state Moore FSM) arms when `sample ≥ 120` on a rising slope, tracks `peak_val` , and confirms a systolic crest when the signal falls by ≥ 25% of `peak_val` .
6	20 ns timestamp	On entering `STATE_PEAK_FOUND` , the FPGA latches a 32-bit 50 MHz cycle counter into `ibi_cycles` , asserts `beat_detected` , stretches a 50 ms pulse onto Blue LED D12, and enters a 250 ms (12,500,000-cycle) refractory blanking state.
7	Full-duplex return	MISO returns `{beat_latched, filt_sample[6:0]}` during the 1-byte sample transfer. When bit 7 is `1` , Core 0 immediately clocks a 5-byte burst (MOSI dummy `0x00`) to read magic byte `0xC3` followed by the big-endian 32-bit cycle count.

#	Pipeline Stage	Hardware / Firmware Execution Detail
8	Mutex hand-off	Core 0 divides cycles by 50,000 to obtain the inter-beat interval (IBI) in milliseconds, validates that it lies within 30–250 BPM (note: the 250 ms hardware refractory window already caps hardware detection at 240 BPM), appends it to the shared ring buffer, and releases `s_data_mutex`.
9	Hazard polling (Core 1)	Core 1 tasks concurrently read the BME280 (1 Hz I²C), PMS5003 (1 Hz UART1), MPU6050 (40 Hz I²C), and GNSS module (1 Hz UART2).
10	SQI & conditioning	Core 1 evaluates the Karlen Signal Quality Index. If $SQI \geq 0.70$, it computes RMSSD and respiration rate (from respiratory sinus arrhythmia). PM2.5 receives its 60 s EPA/AirGradient correction; pressure updates its 180-minute linear regression slope.
11	INT8 TinyML inference	Normalized features `[HR, RMSSD, SpO₂, Temp, RH, PM2.5]` pass through the 619-byte INT8 MLP on the ESP32-S3 FPU in ~20–50 µs to output Heat, Pollution, and Flood/Cold risk scores.
12	Rules override	Deterministic clinical rules (mNEWS2 ≥ 7, SpO_2 < 88%, HR > 150 BPM, or barometric slope ≤ −2.0 hPa/h) override the neural network and force a `CRITICAL` verdict.
13	Local alert & SoftAP	Sustained critical vitals (15 s) or a confirmed fall (10 s grace) move the SOS FSM to `SOS_ACTIVE`: driving the buzzer siren, rendering the full-screen OLED Emergency Card, and serving live JSON at `http://192.168.4.1/api/status`.

3. Dual Hardware Targets & Bus Protocol Architecture

Good silicon engineering separates **proving the digital algorithm** from **optimizing unit cost and power**. VALOR maintains identical, vendor-agnostic core Verilog RTL (`moving_average_8tap.v` and `ppg_peak_detector.v`) across two physical targets:

Architectural Dimension	AMD Xilinx Zynq-7000 Baseline (Golden Reference)	Renesas ForgeFPGA on ShrikeFi (Deployable Demo)
Part & Role	`xc7z020clg400-1` — Formal RTL verification, industrial static timing baseline.	`SLG47910` + `ESP32-S3` — Low-cost, sub-5 mW wearable demonstration platform.
Host Processor	Dual ARM Cortex-A9 hard processing system (PS).	ESP32-S3 dual-core Xtensa LX7 (up to 240 MHz, single-precision FPU).
Interconnect Bus	32-bit memory-mapped AXI4-Lite slave at base address `0x43C00000`.	4-wire 8-bit SPI Mode 0 (`CPOL=0, CPHA=0`) at 1 MHz runtime (16 MHz boot bitstream load).
Sample In / Filtered Out	Write to `REG_IR_RAW`; read 8-bit filtered value from `REG_IR_FILT`.	8-bit raw IR sample on `MOSI`; lower 7 bits returned on `MISO[6:0]` (wraps modulo 128).

Architectural Dimension	AMD Xilinx Zynq-7000 Baseline (Golden Reference)	Renesas ForgeFPGA on ShrikeFi (Deployable Demo)
Beat Flag & IBI Export	`REG_STATUS_THRESH[0]` + hardware interrupt to ARM GIC; read 32-bit `REG_IBI_CYCLES`.	Polled inline on `MISO[7]` (no IRQ wire); followed by 5-byte burst (`0xC3` + 32-bit big-endian cycles).
Arming Threshold	Runtime programmable via `REG_STATUS_THRESH[15:8]`.	Hard-wired at synthesis to `8'd120` to save LUTs (`shrikefi_set_threshold()` caches in RAM only).
Post-Route Resources	185 Slice LUTs, 16 LUTRAMs, 266 FFs, **0 DSP48, 0 BRAM**.	515/1120 LUT5s (45.98%), 303/1856 FFs (16.33%), 105/140 CLBs (75.00%), **0 DSP, 0 BRAM**.
Timing Closure (50 MHz)	OOC WNS **+5.603 ns**, WHS +0.184 ns, F_{max} **69.45 MHz** (Vivado 2026.1 full BD: WNS +12.836 ns / +13.168 ns).	Post-route WNS **+7.316 ns**, TNS 0 ps (1,260 endpoints), F_{max} **78.85 MHz**.
Unit Cost & Verification	$50–$150 SoC, several watts; `tb_ppg_system.v` passes **6/6** self-checking assertions.	~$0.50 FPGA (< 5 mW), ~$12–$15 total BOM; `tb_forgefpga_system.v` passes **13/13** assertions.

Why Each Serial Protocol Was Chosen Across the Board

Protocol	Connected Devices	Wiring & Speed	Engineering Justification
SPI Mode 0	Renesas ForgeFPGA SLG47910	4 wires (`SCK, SS_n, MOSI, MISO`), 1 MHz full-duplex	Master supplies clock (zero baud drift against the FPGA's internal RC oscillator); requires only an 8-bit shift register in RTL; exchanges sample and status simultaneously in 8 µs.
I²C (Shared)	MAX30102, BME280, MPU6050, SSD1306, optional TMP117	2 wires (`SDA, SCL`) with pull-ups, 100 kHz multi-drop	Five low-to-medium-bandwidth peripherals share just two MCU pins using unique 7-bit hardware addresses; total bus utilization is < 15% (MAX30102 uses < 3%).
UART1 & UART2	Plantower PMS5003 (UART1), u-blox NEO-M8N / L89HS (UART2)	2 wires each (`TX, RX`), 9600 baud asynchronous	Both modules are autonomous 1 Hz streamers (32-byte binary frames and NMEA-0183 ASCII lines); ESP32 hardware UART FIFOs buffer them with zero polling overhead.

HISTORICAL TRAP: THE RETIRED 4-BIT "NIBBLE PROTOCOL"

Early migration notes (`MIGRATION_AGENT_TASKS.md`) proposed a 4-bit parallel nibble bus with strobe, direction pins, and a `0x1–0x8` command codebook. It was retired before implementation because it would have consumed 6–8 of the ForgeFPGA's 19 user GPIOs and required CPU bit-banging. The shipped hardware uses pure 4-wire 8-bit SPI on `SPI2_HOST`.

PART II · HARDWARE & RTL

SPI Framing, Clock-Domain Crossing, Verilog Idioms, Filter & Peak-Detector FSM

4. The ESP32-to-FPGA SPI Link & Clock-Domain Crossing (CDC)

On the Vicharak ShrikeFi board, the ESP32-S3 and Renesas ForgeFPGA communicate over internal PCB traces using 4-wire **SPI Mode 0** ($CPOL = 0$: clock idles LOW; $CPHA = 0$: data is sampled on the rising edge and shifted on the falling edge). Two auxiliary pins (`FPGA_EN` and `FPGA_PWR`) are used exclusively at boot when the ESP32 clocks the 46,408-byte SRAM bitstream into the FPGA at 16 MHz. There is no dedicated reset wire and no hardware interrupt line.

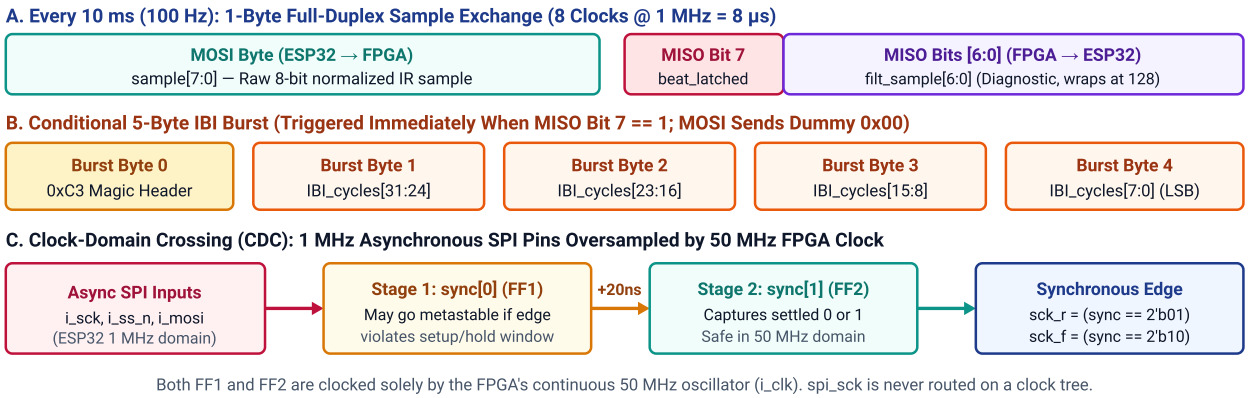


Figure 4.1 — (A) Full-duplex 8-bit sample exchange; (B) 5-byte cycle-accurate IBI burst; (C) 2-FF synchronizer preventing metastability.

Key Protocol Mechanics & Safeguards

- **Why MISO Bits [6:0] Wrap at 128:** Because Bit 7 is reserved for `beat_latched`, only 7 bits remain to return the filtered sample to the MCU, causing values ≥ 128 to wrap around (appearing as a sawtooth if plotted raw). This returned value is strictly for diagnostic telemetry; inside the FPGA, the Peak-Detector FSM receives the full, unclipped 8-bit output `filt_sample[7:0]` directly from the filter.
- **Magic Byte Verification (`0xC3`) & Dummy-MOSI Suppression:** When `beat_latched == 1`, the FPGA loads a 3-bit down-counter (`burst_cnt = 5..1`). While the ESP32 clocks out five dummy `0x00` bytes on MOSI to read the burst, the FPGA suppresses `data_valid` to the filter so the zeros never corrupt the running average. If Burst Byte 0 is not `0xC3` (e.g., due to bus noise), the ESP32 driver discards the burst and falls back to its internal microsecond timer.

- **Metastability & Oversampling Limit:** Because the ESP32's 1 MHz `spi_sck` stops between bytes and is asynchronous to the FPGA's 50 MHz oscillator, wiring `spi_sck` as an FPGA clock net would violate clock-tree rules. Instead, `spi_target.v` oversamples `i_sck`, `i_ss_n`, and `i_mosi` at 50 MHz through 2-flip-flop synchronizers. If FF1 samples an input mid-transition and enters a metastable state, it has a full 20 ns clock period to resolve before FF2 latches it. By Nyquist, a 50 MHz clock cannot sample faster than 25 MHz; accounting for synchronizer delay, the practical ceiling for `spi_sck` is **≤ 5–10 MHz** (1 MHz in normal runtime).

```
// hardware/shrikefi/spi_target.v — 2-FF CDC Synchronizers & Single-Cycle Edge
Detectors
reg [1:0] sck_sync, ss_sync, mosi_sync;
always @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin
        sck_sync <= 2'b00; ss_sync <= 2'b11; mosi_sync <= 2'b00;
    end else begin
        sck_sync  <= {sck_sync[0],  i_sck};
        ss_sync   <= {ss_sync[0],   i_ss_n};
        mosi_sync <= {mosi_sync[0], i_mosi};
    end
end
wire sck_r = (sck_sync == 2'b01); // 1-cycle pulse on rising SCK edge (sample
MOSI)
wire sck_f = (sck_sync == 2'b10); // 1-cycle pulse on falling SCK edge (shift
MISO)
```

5. Seven Verilog RTL Fundamentals Used in VALOR

Writing Verilog is not programming a CPU instruction sequence; it is wiring parallel digital logic gates and flip-flops that execute simultaneously on every 50 MHz clock edge. Every module in VALOR relies on seven foundational RTL constructs:

RTL Concept	How It Appears in VALOR Verilog	Hardware Synthesis Rule & Jury Point
1. Parameterized Module	`module moving_average_8tap #(parameter DATA_WIDTH = 8) (...)`	Allows the exact same RTL file to synthesize for 8-bit, 12-bit, or 16-bit data paths without rewriting logic.
2. Registered Ports (`wire` vs. `reg`)	`input wire clk, spi_sck;` `output reg led_user;`	A `wire` is a continuous point-to-point connection; a `reg` assigned in a clocked `always` block synthesizes to physical D flip-flops driving the pad.
3. State & Counter Flip-Flops	`reg [1:0] current_state, next_state;` `reg [31:0] refractory_cnt, interval_cnt;`	2 flip-flops hold the 4-state binary FSM; 32-bit registers count 20 ns ticks up to 85 seconds without overflow.

RTL Concept	How It Appears in VALOR Verilog	Hardware Synthesis Rule & Jury Point
4. Combinational vs. Sequential	`wire [10:0] next_sum = running_sum + data_in - shift_reg[7];` (comb) `running_sum <= next_sum;` (seq)	Combinational logic updates continuously with propagation delay; sequential logic commits state strictly on `posedge clk`.
5. Two-Block FSM Style	`always @(*)` computes `next_state`; `always @(posedge clk)` updates `current_state <= next_state;`	Separating next-state combinational decoding from state flip-flops prevents accidental transparent latches.
6. Non-Blocking (`<=`) vs. Blocking (`=`)	`shift_reg[0] <= data_in;` `for (i=1; i<8; i=i+1) shift_reg[i] <= shift_reg[i-1];`	**Critical Trap:** Non-blocking `<=` evaluates all right-hand sides before updating any flip-flop. Using blocking `=` here would cascade `data_in` through all 8 stages in one clock and collapse the pipeline into 1 register!
7. Power-On Reset (POR) Counter	`reg [7:0] por_cnt = 8'd0;` `wire rst_n = (por_cnt == 8'd255);`	Holds `rst_n` LOW for the first 255 clocks (5.1 µs) after boot so no external reset pin is needed. (*Note:* submodules currently use `negedge rst_n`; a future revision will dual-flop synchronize deassertion.)

6. The 8-Tap Moving Average Filter (`moving_average_8tap.v`)

Raw photodiode measurements contain thermal noise, 50/60 Hz ambient light flicker, and ADC quantization ripple that would trigger false crests in a derivative detector. VALOR passes every 100 Hz sample through a symmetric 8-tap Finite Impulse Response (FIR) boxcar filter:

$$y[n] = (1/8) \times \sum_{k=0..7} x[n-k] \quad \Rightarrow \quad Sum_{new} = Sum_{old} + x[n] - x[n-8]$$

Summing eight 8-bit values from scratch every cycle would require a multi-level adder tree. Instead, the filter keeps an 11-bit `running_sum`: when `data_valid` pulses, it adds the newest sample `data_in` and subtracts the oldest sample `shift_reg[7]` (which represents $x[n-8]$) leaving the 8-stage shift register.

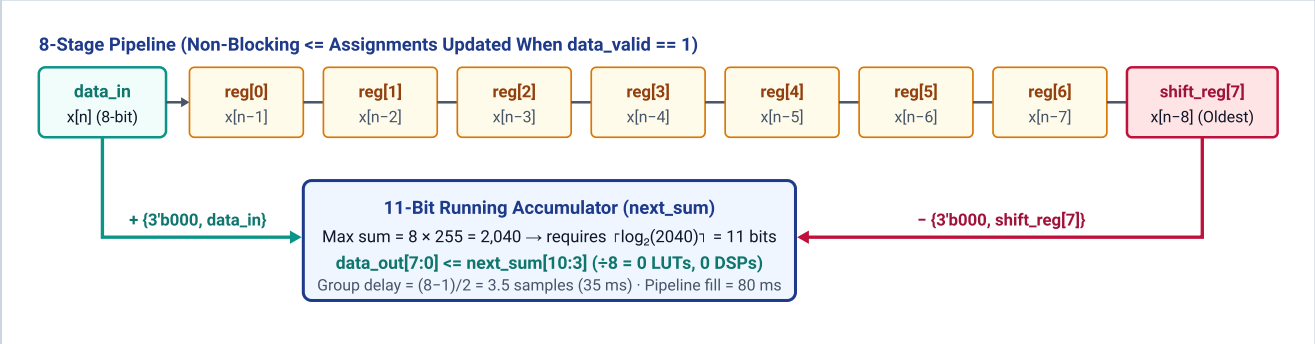


Figure 6.1 — Datapath of `moving_average_8tap.v`. Dividing by 8 is a pure wire routing shift (`next_sum[10:3]`) that consumes zero LUTs.

- **Bit-Width Growth Math:** Eight maximum 8-bit inputs ($8 \times 255 = 2{,}040$) require $8 + log_2(8) = 11$ bits (`[10:0]`) to guarantee zero arithmetic overflow. For any generic `DATA_WIDTH`, the accumulator width is `DATA_WIDTH + 3`.
- **Why Dividing by 8 Is Free in Hardware:** Dividing by $2^3 = 8$ in binary is a 3-bit right shift. In an FPGA, a constant shift requires no logic gates at all—bit 3 of `next_sum` is wired directly to bit 0 of `data_out`, bit 4 to bit 1, up to bit 10 to bit 7. Dividing by a non-power of two (such as 7) would require a multi-LUT divider circuit.
- **Why 8 Taps Instead of 16 or 32:** At 100 Hz, an 8-tap FIR filter has a symmetric group delay of $(8 - 1) / 2 = 3.5$ samples (**35 ms**) and fills in **80 ms**. Expanding to 16 or 32 taps would flatten the sharp systolic crest needed for peak timing and double or quadruple flip-flop usage on the 1,120-LUT ForgeFPGA.

7. Peak-Detector FSM, Synthesis Utilization & Timing Closure

`hardware/common/ppg_peak_detector.v` is the heart of the FPGA accelerator. It is a **registered 4-state Moore finite state machine** encoded in **2-bit binary** (`00, 01, 10, 11`) to save flip-flops compared to 4-bit one-hot encoding. Because it is a pure Moore machine, `beat_detected` depends strictly on `current_state == STATE_PEAK_FOUND` inside a clocked block, guaranteeing a glitch-free 1-cycle output pulse.

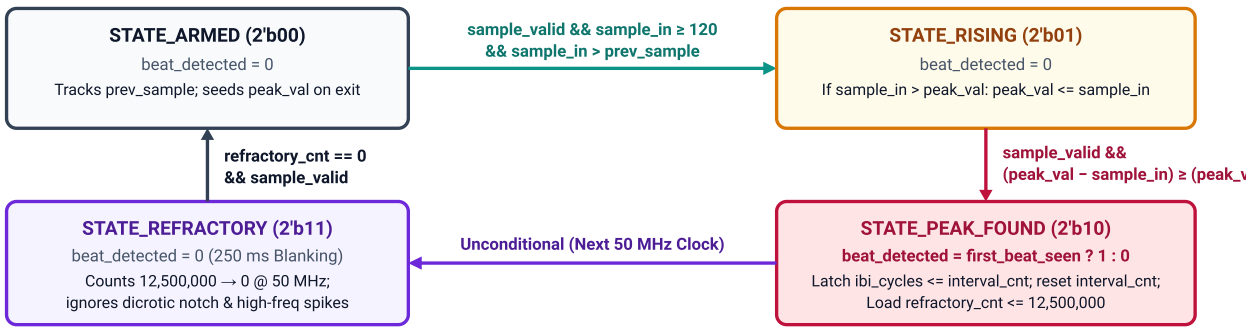


Figure 7.1 — State-transition diagram of `ppg_peak_detector.v`. Binary 2-bit encoding uses 2 flip-flops; Moore outputs prevent combinational glitches.

- **Proportional Crest-Fall Rule & Unreachable Branch Disclosure:** The RTL defines `wire [7:0] crest_fall_min = (peak_val > 8'd48) ? (peak_val >> 2) : 8'd6;`. Because entering `STATE_RISING` requires `sample_in ≥ 120`, `peak_val` is always ≥ 120 in this build. Consequently, the **25% drop branch (`peak_val >> 2`) is always active**, and the `8'd6` weak-signal floor is unreachable hardware. Volunteering this detail shows deep familiarity with the synthesized netlist.

- **First-Beat Suppression (`first_beat_seen`):** On the very first pulse after reset, `interval_cnt` has been counting since power-on rather than from a prior heartbeat. The register `first_beat_seen` suppresses `beat_detected` on that first peak so it silently seeds the interval timer without emitting a bogus first IBI or LED flash.
- **250 ms Refractory Window Math:** After aortic valve closure, the arterial pulse exhibits a secondary bounce called the **dicrotic notch**. Blanking the detector for $0.250\text{ s} \times 50{,}000{,}000\text{ Hz} = 12{,}500{,}000$ clock cycles prevents double-counting the dicrotic notch while supporting heart rates up to $60\text{ s} \div 0.250\text{ s} = 240\text{ BPM}$.
- **Worked RMSSD Example & Why 20 ns Matters:** Given five consecutive IBIs of **820, 840, 800, 860, and 830 ms** (corresponding to 41.0M, 42.0M, 40.0M, 43.0M, and 41.5M FPGA cycles), the four successive differences are $+20, -40, +60, -30\text{ ms}$. Squaring ($400, 1600, 3600, 900$), averaging ($6500 \div 4 = 1625$), and taking the square root yields $RMSSD = \sqrt{1625} \approx 40.31\text{ ms}$. Because healthy resting RMSSD is 10–50 ms, a 5 ms FreeRTOS scheduling tick jitter injects a **10% to 50% error** directly into HRV.

ForgeFPGA SLG47910 Post-Route Resource Utilization & LUT Evolution

In the production bitstream (`6930ba88...` , report `resource_utilization_spi_link.log`), **CLB occupancy (75.00%) is the true binding constraint**, not LUT count (45.98%). Each Configurable Logic Block holds 8 LUT5s and flip-flops; wide 32-bit counters and shift registers span across CLB slices even when individual LUTs inside those slices remain unmapped.

SLG47910 Resource	Used / Available	Utilization	Architectural Role & Headroom
CLB LUT5s	515 / 1,120	45.98%	8-tap filter, Peak-Detector FSM, 32-bit counters, and 5-byte IBI burst muxes (605 LUTs free).
Flip-Flops (FF)	303 / 1,856	16.33%	298 CLB FFs + 5 I/O FFs: shift pipeline, state registers, 32-bit timers, 2-FF CDC synchronizers.
CLBs (Slices)	**105 / 140**	**75.00%**	**Binding floorplan limit** driven by wide 32-bit counter routing across CLB tiles.
User GPIOs	5 / 19	26.32%	`spi_sck` , `spi_ss_n` , `spi_mosi` , `spi_miso` , and 50 ms stretched `led_user` (D12).
4k Block RAM & PLLs	0 / 8 BRAM · 0 / 1 PLL	0.00%	Distributed flip-flops used instead of SRAM; clocked directly from the 1/1 internal 50 MHz oscillator.

Why LUTs Grew from 342 to 515 (Deliberate Capability Upgrade): Early decks quoted 342 LUTs (Sep 16: filter + basic comparator, 194 FFs, 76 CLBs). On Sep 21 (363 LUTs, 202 FFs, 75 CLBs) and Oct 17 (362 LUTs, 204 FFs, 74 CLBs), the positive-slope arming guard was added. On **Oct 18 (535 LUTs, 303 FFs, 108 CLBs)**, we added the 32-bit 20 ns IBI counter, the 5-byte `0xC3` burst serializer, and dummy-MOSI suppression. Finally, on **Oct 19–20 (515 LUTs, 303 FFs, 105 CLBs)**, the crest drop was raised from `peak/8` to `peak/4` to reject the dicrotic notch and post-route timing closed at 50 MHz with **WNS +7.316 ns** (12.684 ns critical path vs. 20.000 ns deadline; F_{max} 78.85 MHz).

PART III · SOFTWARE & INTELLIGENCE

ESP32-S3 Dual-Core Firmware, INT8 TinyML, Clinical Rules & Sensors

8. ESP32-S3 Dual-Core Firmware & Offline SoftAP Architecture

The firmware is written in C11 on Espressif ESP-IDF v5.5.x using FreeRTOS preemptive priority scheduling (tick rate `CONFIG_FREERTOS_HZ = 200` → 5.0 ms scheduling quantum). To guarantee that Wi-Fi bursts, HTTP requests, or neural network inference never delay optical sampling, tasks are strictly pinned across the two Xtensa LX7 cores and decoupled via the FreeRTOS mutex `s_data_mutex`.

FreeRTOS Task	Core	Pri	Stack	Rate	Dedicated Responsibility
`task_ppg_accelerator`	**Core 0**	5 (Max)	7 KB	100 Hz	Reads MAX30102 FIFO (`INT_ENABLE_1 = 0x40` , bit 6 `PPG_RDY`), drives 1 MHz SPI to FPGA, captures 5-byte IBI burst, computes SpO_2 ratio-of-ratios.
`task_disaster_monitor`	**Core 1**	2	10 KB	1 Hz	Master clinical loop: snapshots shared vitals under `s_data_mutex` , polls BME280, runs Karlen SQI, 180-min pressure regression, INT8 TinyML, mNEWS2, SOS FSM, and flushes the 1,024-byte SSD1306 OLED buffer.
`task_pms5003_uart`	**Core 1**	3	4 KB	1 Hz	Parses 32-byte UART1 packets, verifies 16-bit checksum, feeds 60 s EPA/AirGradient rolling calibration window.
`task_gps_uart`	**Core 1**	3	4 KB	1 Hz	Parses UART2 NMEA-0183 sentences into latitude/longitude or triggers indoor NVS fallback.
`task_imu_monitor`	**Core 1**	3	4 KB	40 Hz	Polls MPU6050 over I²C, runs 5-state Fall Detection FSM, computes motion energy for PPG gating.

FreeRTOS Task	Core	Pri	Stack	Rate	Dedicated Responsibility
`Wi-Fi + HTTP Daemon`	**Core 1**	lwIP	—	Event	Pinned via `CONFIG_ESP_WIFI_TASK_PINNED_TO_CORE_1=y`. Broadcasts open SSID `VALOR-0081` and serves `valor_dashboard.html` from `.rodata` flash at `http://192.168.4.1` (endpoints: `/`, `/api/status`, `/status.json`, `POST` location/cancel).

- **Preventing Torn Reads & I²C Bus Lockups:** Because 32-bit floating-point structures are not updated atomically across two cores, Core 0 locks `s_data_mutex` for a few microseconds to write new IBIs, and Core 1 locks it briefly at 1 Hz to copy a local snapshot. On the shared I²C bus, `esp32_i2c_hal.c` enforces a hard **50 ms transaction timeout**: if a jumper glitches, the HAL resets the I²C peripheral controller and returns an error code rather than hanging the RTOS task. Stack overflow protection is enabled via `CONFIG_FREERTOS_CHECK_STACK_OVERFLOW`.

9. The Edge AI Layer: 619-Byte INT8 TinyML & MIMIC-III Validation

VALOR executes a feed-forward Multilayer Perceptron (MLP) of topology **6 → 24 → 16 → 3** directly on the wearable in **20–50 µs** (measured at the 160 MHz CPU clock setting; 0.44 µs on an x86 host), compared to 200–2,000 ms for a cloud round-trip. The six min-max normalized inputs `[HR, RMSSD, SpO₂, Ambient Temp, Humidity, PM2.5]` pass through two hidden ReLU layers ($f(x) = max(0, x)$) to produce three independent sigmoid risk scores ($0.0 \text{ to } 1.0$): **Heat Risk**, **Pollution Risk**, and **Flood/Cold Risk**.

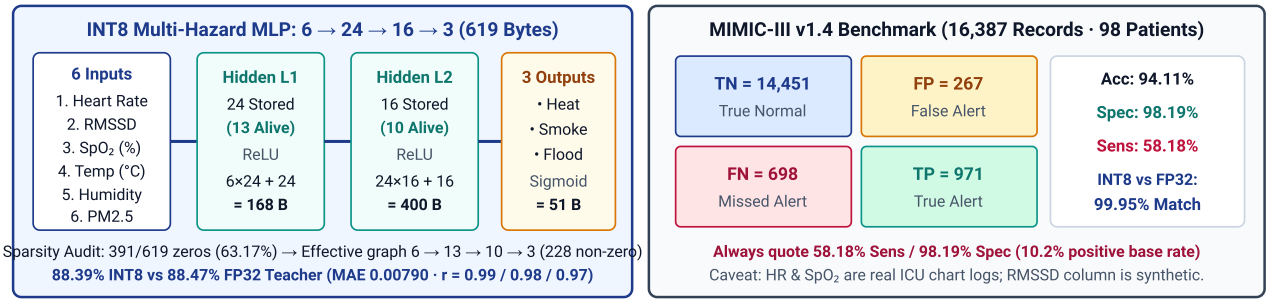


Figure 9.1 — Left: Stored vs. effective topology of the 619-byte INT8 risk network. Right: MIMIC-III v1.4 evaluation across 16,387 ICU records.

- **On-the-Fly FPU Dequantization Kernel:** The 576 weights and 43 biases ($168 + 400 + 51 = 619 \text{ parameters}$) are stored in flash as signed 8-bit integers (`619 bytes` vs. in float32, a 75% reduction). During inference (), each weight is dequantized on the fly using a per-layer scale factor () and evaluated in 32-bit floating point on the ESP32-S3 hardware FPU (576 MACs + 3 sigmoids). Across 7,000 held-out test vectors, the INT8 model achieves accuracy vs. for the float32 teacher (97.32% risk-tier agreement, MAE 0.00790, max error 0.09136).

- `2,476 bytes` `AI_MODEL_AUDIT_2026-10-20.md` $w_{float} = w_{int8} \times S$ `expf()` **88.39%88.47%**
- **Honest Sparsity Audit (`nn_risk_model_int8.c.inc`):** Exactly 391 of the 619 stored INT8 parameters (63.17%) are zero. Specifically, 11 of 24 Layer-1 neurons and 6 of 16 Layer-2 neurons have all-zero weights and negative biases (always outputting 0 after ReLU), making the active compute graph an effective **6 → 13 → 10 → 3** network with 228 non-zero parameters.
- **Why We Deleted the 3 → 8 → 4 → 1 PM2.5 Neural Calibrator:** An early build used a small neural network (and a 32-step LUT sigmoid) to calibrate raw PMS5003 readings against temperature and humidity. Bench audits on 2026-10-20 revealed that it suppressed dry-air readings by 31–53% and collapsed low-concentration smoke readings to `0 µg/m³` in high humidity. In a disaster zone, telling a firefighter that toxic air is clean is a fatal failure mode. We deleted the neural calibrator and replaced it with the peer-reviewed **US EPA / AirGradient banded polynomial equations** (`pm25_correction.c`) over a 60-second rolling window with explicit confidence flags (`PM25_CONF_OK`, `HUMID`, `DUST`, `SETTLING`).

10. Clinical Triage Rules, Cyclone Regression, Fall & SOS FSMs

Neural networks are statistical approximators that must never be trusted alone in life-critical situations. VALOR wraps the TinyML model inside a **deterministic clinical safety envelope** that always supersedes the AI score during an emergency.

Severity Tier	mNEWS2 Clinical Score	Heat Strain Ladder (Any One)	Toxic Smoke Ladder (Any One)
Caution	Score 0–4 (Low clinical risk; routine monitoring)	Heat Index ≥ 35 °C	PM2.5 ≥ 35 µg/m³ (EPA 24h std; WHO is 15) or SpO_2 ≤ 96%
Moderate	Score 5–6 (Medium risk; urgent clinical review)	HI ≥ 40 °C or HR ≥ 95 BPM or RMSSD ≤ 35 ms	PM2.5 ≥ 75 µg/m³ or SpO_2 ≤ 94% (or EPA AQI 151–200 floor)
High	Any single parameter scoring 3 points (Red Flag, e.g., SpO_2 ≤ 91%)	HI ≥ 45 °C or HR ≥ 110 BPM or RMSSD ≤ 20 ms	PM2.5 ≥ 150 µg/m³ or SpO_2 ≤ 92% or HR ≥ 100 BPM (or AQI 201–300)
CRITICAL (Hard Override)	**mNEWS2 ≥ 7** (or forced override: **SpO_2 < 88%** or **HR > 150 BPM**)	**HI ≥ 54 °C** or **HR ≥ 130 BPM** or **RMSSD ≤ 10 ms**	**PM2.5 ≥ 300 µg/m³** or **SpO_2 ≤ 88%** or **HR ≥ 120 BPM** (or AQI 301+)

Barometric Cyclone Engine (`pressure_trend.c`): Because absolute barometric pressure varies with elevation (a mountain town sits far below 1000 hPa on a sunny day), VALOR computes an ordinary least-squares linear regression slope over a **180-minute rolling window** of BME280 readings. To prevent sensor drift or elevator rides from false-alarming, a noise floor requires a minimum total drop of **0.5 hPa** before evaluating slope thresholds: **≤ −0.5 hPa/h** (1.5 hPa/3h) = Moderate weather change; **≤ −1.0 hPa/h** (3.0 hPa/3h) = High storm/gale warning; **≤ −2.0 hPa/h** (6.0 hPa/3h) = **CRITICAL cyclone precursor** (forces overall CRITICAL triage override).

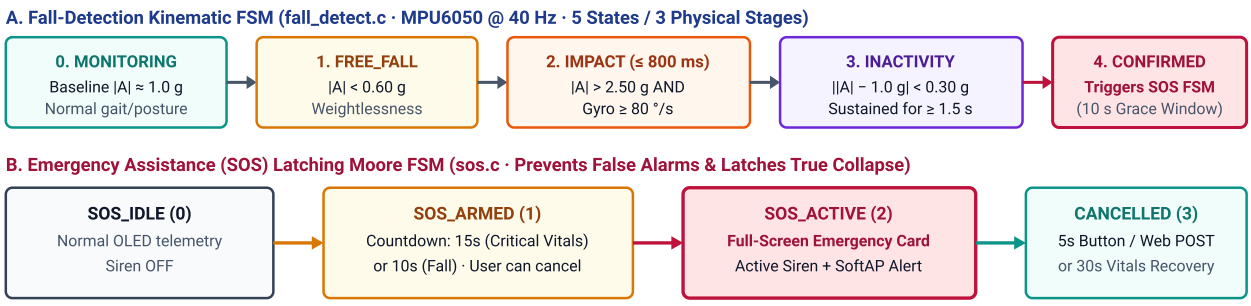


Figure 10.1 — (A) Fall-Detection kinematic sequence rejecting dropped bags; (B) SOS escalation Moore FSM with grace and recovery timers.

11. Sensor Physics, Two-Source Skin Temperature & Indoor Location

- **SpO_2 Ratio-of-Ratios (`MAX30102`):** Oxygenated hemoglobin absorbs more infrared light (880 nm), whereas deoxygenated hemoglobin absorbs more red light (660 nm). Separating the pulsatile arterial AC component from the steady tissue/venous DC baseline yields $R = (AC_{Red} / DC_{Red}) \div (AC_{IR} / DC_{IR})$, mapped via the empirical bench fit $SpO_2 \approx 101.8 - 6.67 \times R$ (calibrated LED drive current: static 11.0 mA / register `0x37`, ceiling 18.0 mA / `0x5A`).
- **Two-Source Skin Temperature & The Measured +4.39 °C Offset:** VALOR supports both an optional high-precision TI `TMP117` contact probe ($\pm 0.0078125 \text{ °C/LSB}$, auto-probed at `0x48–0x4B` at boot) and the always-present `MAX30102` silicon die sensor. Every payload in `/status.json` tags `skinsrc` as `NONE`, `ESTIMATED` (displayed as `Skin ≈ XX.X °C`), or `TMP117` (displayed as `Skin XX.XX °C (TMP117 probe)`). On the bench, we measured the MAX30102 LED self-heating offset directly: **28.44 °C die vs. 24.05 °C room ambient = +4.39 °C** (exposing an earlier bug where a guessed `+5.5 °C` guard sat above the 30.07 °C peak and forced the estimator to output 0.00 °C). Rather than using a static linear formula, `skin_estimate.c` freezes the pre-contact idle baseline upon finger contact and evaluates the relative thermal step:

$$T_{skin} = 34.10 + 0.55 \times (T_{die_pred} - T_{die_baseline}) + 0.12 \times (T_{amb} - 27.5)$$

  with a 0.28 °C/s slew limit and 5-sample median filter, publishing 3 s after contact and settling at 12 s. Against a 36.10 °C reference, the nominal model reads 34.87 °C (a documented **−1.23 °C gap**). Because a peripheral fingertip naturally sits at 30–34 °C, feeding it into the core-temperature NEWS2 table would falsely score +3 hypothermia on every healthy person; therefore, **mNEWS2 temperature points are held at 0**.
- **Three-Tier Location Fallback (`gnss_nmea.c`, `location.c`):** While the u-blox NEO-M8N / Quectel L89HS GNSS driver passes 12/12 unit tests and locks outdoors, 1.5 GHz satellite signals cannot penetrate concrete shelters or storm debris and draw 15–25 mA continuously. When no satellite fix is present, VALOR automatically falls back to a human-readable deployment label stored in non-volatile flash (e.g., *"Disaster Shelter 4, Ward B"*), which any rescuer can update offline via HTTP POST from the `VALOR-0081` dashboard.

PART IV · PROVE & PRESENT

Verification Matrix, 5-Act Demo Runbook, Limitations & Reconciliations

12. End-to-End Verification & Evidence Matrix

Verification Domain	Tool / Test Harness	Quantitative Evidence	Honest Status
Zynq-7000 Baseline	Vivado ML 2022.2 (OOC) & 2026.1 (Full BD)	185 LUT, 16 LUTRAM, 266 FF, 0 DSP, 0 BRAM; OOC WNS +5.603 ns (69.45 MHz); Full BD WNS +12.836 ns / +13.168 ns.	Verified golden reference
ForgeFPGA Fit & Timing	Renesas Fitter + 50 MHz SDC Post-Route	515/1120 LUT5 (45.98%), 303/1856 FF (16.33%), 105/140 CLB (75%), 0 BRAM; WNS +7.316 ns, TNS 0 ps, F_{max} 78.85 MHz.	Production bitstream `6930ba88...`
Verilog RTL Simulation	Icarus Verilog (`iverilog + vvp`)	`tb_forgefpga_system.v` : **13/13** assertions pass; `tb_ppg_system.v` : **6/6** assertions pass.	100% passing (runnable live)
MIMIC-III Clinical Harness	`mimic_harness.c` & `accuracy_evaluator.c`	16,387 ICU records (98 patients): 94.11% acc, 98.19% spec, 58.18% sens; 99.95% (16,379/16,387) FP32-to-INT8 match.	Validated (synthetic RMSSD caveat)
INT8 Quantization Audit	GCC 15.2 Host Harness (7,000 samples)	88.39% INT8 vs. 88.47% FP32 teacher; 97.32% tier match; MAE 0.00790, max error 0.09136.	Fully verified
Clinical & Hazard Rules	`test_disaster_risk_engine.c`	34 boundary-sweep unit tests covering mNEWS2, Rothfusz Heat Index, PM2.5 EPA curves, and 180-min pressure slope.	100% passing (34/34)
Fall Detection & TMP117	Host FSM Suites + Bench Drop	16/16 Fall FSM tests + 1/1 physical drop test (2026-10-19); 16/16 TMP117 driver tests (one physical breakout dead on arrival).	Verified in code & bench
GNSS NMEA Parser	`test_gps_l89hs.c`	12/12 unit tests (checksum verification, hemisphere signs, corrupted-frame rejection).	100% passing (12/12)

13. The 7-Minute, 5-Act Live Demonstration Runbook

Act & Window	Physical Action on Stage	Key Engineering Proof & Spoken Line
Act 1 (0:00–1:00) Offline Boot	Hold up unpowered ShrikeFi board; point out ForgeFPGA and ESP32-S3. Plug in USB-C; show OLED booting to `UNKNOWN ("Patient not assessed")`.	**Proof:** Refuses to fabricate vitals before contact. *"In a disaster the cloud dies first. VALOR pairs a sub-dollar 50 MHz FPGA for 20 ns pulse timing with a dual-core ESP32-S3 for INT8 TinyML triage, 100% offline."*
Act 2 (1:00–2:30) FPGA PPG Lock	Place finger steadily on MAX30102. Point to Blue LED D12 flashing on every pulse and OLED showing HR (~74 BPM) & SpO_2 (~98%). Launch 60 FPS Web-Serial scope.	**Proof:** Blue LED D12 is driven directly by an FPGA pin (holding FPGA in reset stops D12 while ESP32 keeps logging). *"RTOS scheduling adds 5 ms jitter that ruins HRV; our FPGA timestamps crests to 20 ns."*
Act 3 (2:30–4:00) Hazard Fusion	Cup hand over BME280 to raise temperature/humidity; point to 180-minute barometric slope gauge and EPA-corrected PM2.5 confidence card (`PM25_CONF_OK`).	**Proof:** Multi-parameter fusion prevents alarm fatigue. *"We track 3-hour barometric slope rather than raw pressure, and use EPA polynomial smoke calibration with explicit confidence flags."*
Act 4 (4:00–5:30) Fall → SOS Card	Drop board ~15 cm onto a notebook and leave still for 1.5 s. Show `*** FALL DETECTED ***` 10 s countdown expiring into the full-screen `MEDICAL EMERGENCY` card + siren.	**Proof:** 5-state kinematic sequence (`FREE_FALL → IMPACT → INACTIVITY`) + 10 s grace timer. *"A dropped bag keeps moving or gets picked up; true incapacitation is impact followed by stillness."*
Act 5 (5:30–7:00) Rescuer SoftAP	On a smartphone (with mobile data off), join Wi-Fi `VALOR-0081` and open `http://192.168.4.1`. Show live alert and tap `CANCEL SOS` from the phone browser.	**Proof:** Zero app install, zero router, zero internet. *"The wearable compiles its entire HTML/JS dashboard into flash and serves any rescuer's phone over local radio."*

Stage Failure Recovery Protocol: (1) *Permanent* `NO_FINGER` *on MAX30102:* Overhead fluorescent lighting is saturating the photodiode—cup your hand over the sensor. (2) *OLED stays dark:* Check 3.3 V / I²C jumpers and power-cycle USB-C; the 50 ms HAL timeout recovers the bus automatically. (3) *Phone drops Wi-Fi connection:* Turn off cellular data on the phone so the OS does not auto-disconnect from an internet-less hotspot. (4) *Total hardware fault:* Open `valor_dashboard.html` in Chrome Simulation Mode (*Heat Wave*, *Severe Smog*, *Flash Flood* profiles) and run `iverilog` + `vvp` live in terminal to prove all 13 RTL assertions.

14. Honest Limitations & Unified Technical Reconciliations

What VALOR Is vs. What VALOR Is Not: VALOR is an **offline edge engineering prototype for multi-hazard physiological screening and rescue triage**. It is **not** an FDA 510(k) or CDSCO-certified medical diagnostic monitor, nor a replacement for a 12-lead hospital ECG (clinical approval requires ISO 13485 manufacturing, IEC 60601-1 electrical isolation, and prospective human hypoxia trials).

Master Reconciliation Table (Resolving All Historical & Editing Discrepancies)

Specification Topic	Apparent Discrepancy Across Early Docs	Authoritative Reconciled Engineering Fact
1. HRV Settling Window (30 vs. 240 Beats)	Mentioned as "30 valid beats (~20–30 s)" in demo notes vs. "240 beats (~3 min, `settling n/240`)" in clinical Q&A.	Both exist in firmware: the ring buffer computes its first preliminary RMSSD after 30 valid beats, but the UI shows `settling n/240` until 240 beats (~3 min) accumulate for clinical-grade HRV stability.
2. Battery Life & Current Draw	1,200 mAh LiPo → 10–14 h at 80–120 mA average vs. 500 mAh LiPo → 8–12 h duty-cycled (while Wi-Fi ~80 mA + PMS5003 fan ~100 mA = 180 mA peak).	Continuous peak with PMS5003 fan and Wi-Fi TX is ~180 mA. Duty-cycling the PMS5003 fan and Wi-Fi beacon brings average draw to 80–120 mA (10–14 h on a 1,200 mAh pouch; a 500 mAh cell requires deeper sleep duty-cycling).
3. ESP32-S3 Clock (240 vs. 160 MHz)	Architecture diagrams state 240 MHz max clock; TinyML benchmark quotes 160 MHz.	The Xtensa LX7 silicon is rated to 240 MHz; the 20–50 µs INT8 inference benchmark was profiled at the power-saving 160 MHz clock configuration.
4. Upper Heart-Rate Bound (240 vs. 250 BPM)	FPGA refractory math states 240 BPM ceiling; C firmware sanity check accepts 30–250 BPM.	The FPGA's 250 ms refractory window sets the binding hardware ceiling at **240 BPM**; the firmware's 240 ms (250 BPM) software filter is a secondary sanity check.
5. Fall FSM (3-Stage vs. 5-State)	Described as "3-stage fall detection" in prose and a "5-state FSM" in code.	Three physical kinematic stages (`FREE_FALL`, `IMPACT`, `INACTIVITY`) sit between two bookend states (`MONITORING` and `CONFIRMED`) = 5 states total.
6. Peak FSM Names & LUT Counts	Early slides listed `IDLE/ARMED/CREST_FALL/REFRACTORY` (drop=3) and 342 LUTs.	Production Verilog (`ppg_peak_detector.v`) uses `ARMED/RISING/PEAK_FOUND/REFRACTORY` (drop=25%) and synthesizes to **515/1120 LUT5s** (105/140 CLBs) after adding the 32-bit IBI burst.

PART V · JURY DEFENSE & REFERENCE

Consolidated 30-Question Judge Playbook, Team Roles, Quick Card & Glossary

15. The 30-Question Jury Defense Playbook

Every question below includes a crisp **20-Second Answer**, a deeper **Technical Follow-Up**, and the **Credibility Trap ("Don't Say")** to avoid.

#	Judge Question	20-Second Answer & Technical Follow-Up	Trap ("Don't Say")
1	Why use an FPGA if the ESP32-S3 runs at 240 MHz?	**A:** FreeRTOS preemptive scheduling adds ~5 ms of tick jitter (`CONFIG_FREERTOS_HZ = 200`). Because healthy RMSSD HRV is 10–50 ms, 5 ms of OS jitter injects a 10–50% error. The FPGA timestamps peaks in dedicated hardware at 50 MHz with **20 ns resolution**, immune to CPU load.	*"The ESP32 is too slow to filter."* (The issue is deterministic timing jitter, not raw clock speed.)
2	How is VALOR different from an Apple Watch or Garmin?	**A:** Consumer watches rely on cloud APIs, have no PM2.5 smoke sensor or 3-hour barometric cyclone regression, suffer OS scheduling jitter, and cannot broadcast an open zero-install Wi-Fi triage dashboard to rescue teams.	*"Smartwatches are poorly made."* (Frame it as consumer fitness vs. offline disaster triage.)
3	How does VALOR operate with zero internet?	**A:** All signal conditioning, the 619-byte INT8 neural network, mNEWS2 rules, and `valor_dashboard.html` are compiled into internal flash. The ESP32 hosts open SoftAP `VALOR-0081` at `192.168.4.1` over local radio.	*"It uses satellite internet."* (It uses passive GNSS reception only; zero external backhaul.)
4	Is your Peak-Detector FSM a Moore or Mealy machine?	**A:** A **registered Moore machine**: `beat_detected` is assigned inside a clocked block strictly from `current_state == STATE_PEAK_FOUND` , never combinationally from `sample_in` , ensuring a glitch-free 1-cycle pulse.	*"Mealy, because inputs trigger transitions."* (Moore vs. Mealy defines how outputs are generated.)
5	Why 2-bit binary state encoding instead of one-hot?	**A:** Binary encoding uses 2 flip-flops for 4 states (`00..11`), whereas one-hot requires 4 flip-flops. On a compact 1,120-LUT ForgeFPGA where CLB slices are 75% utilized, conserving flip-flops and routing is priority.	*"Binary decoding is faster."* (One-hot is slightly faster to decode in large FPGAs.)
6	How do you handle Clock-Domain Crossing (CDC) on SPI?	**A:** `spi_sck` is never used as a clock net because it stops between bytes. Instead, the FPGA oversamples `i_sck` , `i_ss_n` , and `i_mosi` at 50 MHz through **2-flip-flop synchronizers** (giving FF1 20 ns to settle) and detects edges synchronously (`sck_r` , `sck_f`).	*"We wired SPI SCK to the FPGA clock pin."* (Gated clocks violate FPGA timing analysis.)
7	What is WNS and what timing margin did you achieve?	**A:** Slack is $T_{required} - T_{arrival}$; Worst Negative Slack (WNS) is the margin of the slowest path. At 50 MHz (20.0 ns period), ForgeFPGA closed post-route at **WNS +7.316 ns** (TNS 0 ps, F_{max} 78.85 MHz) and Zynq OOC at **+5.603 ns** (69.45 MHz).	*"A small negative slack is okay."* (Any negative slack violates setup time and causes metastability.)
8	Why is BRAM 0% and what is the real FPGA bottleneck?	**A:** An 8-tap pipeline and 4-state FSM map cleanly into distributed flip-flops (303 FFs), leaving all 8 Block RAMs free. The binding constraint is **CLB tile occupancy at 75.00% (105/140 CLBs)** due to wide 32-bit counters, while LUT5s sit at 45.98% (515/1120).	*"The FPGA ran out of LUTs or has no BRAM."* (It has 8 BRAMs and 605 free LUT5s.)

#	Judge Question	20-Second Answer & Technical Follow-Up	Trap ("Don't Say")
9	How does dividing by 8 in the filter cost zero logic gates?	**A:** Dividing by $2^3 = 8$ is a 3-bit right shift. In Verilog (`data_out <= next_sum[10:3]`), wires from bits 3..10 of the 11-bit running sum connect directly to bits 0..7 of the output—consuming 0 LUTs and 0 DSPs.	*"The compiler optimizes the division instruction."* (There is no CPU or instruction stream in RTL.)
10	What is the exact SPI frame format between MCU and FPGA?	**A:** Every 10 ms, 1 byte is exchanged full-duplex at 1 MHz: 8-bit raw IR on MOSI, and `{beat_latched, filt_sample[6:0]}` back on MISO. When bit 7 is `1` , 5 extra bytes follow: magic header `0xC3` + big-endian 32-bit IBI cycle count.	*"It sends a multi-byte command header every sample."* (The 1-byte sample transfer is self-framing.)
11	Why was the 4-bit parallel nibble protocol retired?	**A:** It would have consumed 6–8 of the FPGA's 19 GPIOs and required software bit-banging. Hardware SPI Mode 0 uses 4 pins and completes a byte in 8 µs (0.08% bus duty cycle).	*"We still use the nibble bus."* (It was retired in design review and never synthesized.)
12	What happens if the ESP32 runs SPI SCK at 20–30 MHz?	**A:** Communication fails. Because the FPGA oversamples SCK with a 50 MHz clock, the Nyquist limit is 25 MHz, and with a 2-FF synchronizer SCK must stay **≤ 5–10 MHz** so rising/falling edges are never missed.	*"The PCB traces overheat."* (It is a synchronous Nyquist oversampling limit.)
13	What is the status of the TMP117 temperature probe?	**A:** The driver (16/16 unit tests passing) auto-probes `0x48–0x4B` at boot. One physical breakout module failed to ACK on a healthy 3.18 V I²C bus, so the firmware seamlessly falls back to the MAX30102 die estimator and tags `skinsrc` in JSON.	*"We abandoned contact temperature sensing."* (The auto-switching driver is shipped and active.)
14	Why not score body fever from the MAX30102 die sensor?	**A:** The die sits under LEDs that self-heat it by a measured **+4.39 °C** above room air, and a fingertip is a peripheral site (30–34 °C), not core body temperature. We compute a baseline-relative skin estimate (labelled `≈` , with a known −1.23 °C gap) and hold mNEWS2 temperature points at 0.	*"The MAX30102 measures core body temperature."* (Never claim core temperature from a fingertip.)
15	How does the cyclone engine avoid altitude false alarms?	**A:** It ignores absolute pressure and fits a least-squares linear regression slope over a **180-minute rolling window**, alerting at ≤ −1.0 hPa/h (storm) and ≤ −2.0 hPa/h (critical cyclone) only after a ≥ 0.5 hPa total drop.	*"It triggers an alarm below 1000 hPa."* (Any high-altitude city sits below 1000 hPa daily.)
16	Does your TinyML neural network diagnose medical diseases?	**A:** No. It computes continuous multi-hazard physiological stress scores (Heat, Smoke, Flood/Cold). Life-critical alarm escalation is governed by deterministic mNEWS2 and hard vital overrides.	*"Our AI diagnoses sepsis or heart attacks."* (It is a hazard-triage screening aid.)

#	Judge Question	20-Second Answer & Technical Follow-Up	Trap ("Don't Say")
17	What does 94.11% accuracy on MIMIC-III actually prove?	**A:** Across 16,387 ICU records (98 patients), our embedded C pipeline and INT8 model reproduce our clinical triage rules with **58.18% sensitivity / 98.19% specificity** (94.11% accuracy) and **99.95% FP32-to-INT8 decision agreement**.	*"It is 94.11% accurate at replacing ICU doctors."* (Always state the 10.2% base rate and synthetic RMSSD caveat.)
18	Why did you delete the 3 → 8 → 4 → 1 PM2.5 neural network?	**A:** Bench testing on 2026-10-20 showed it cut dry-air PM2.5 readings by 31–53% and collapsed low smoke readings to zero in high humidity. We replaced it with published US EPA / AirGradient polynomial equations and explicit confidence flags.	*"It didn't fit in flash memory."* (It was only a few dozen bytes; we removed it because it was unsafe.)
19	Why store weights in INT8 if the ESP32-S3 has a float FPU?	**A:** Storing 619 parameters as signed INT8 cuts flash footprint by 75% (619 B vs. 2,476 B) while on-the-fly FPU dequantization ($w_{float} = w_{int8} \times S$) avoids fixed-point saturation and loses only 0.08% accuracy (88.39% vs. 88.47%).	*"Quantization makes the neural network more accurate."* (It trades 0.08% accuracy for 75% memory savings.)
20	How do you prevent race conditions between Core 0 and Core 1?	**A:** Shared vitals and the IBI ring buffer are guarded by the FreeRTOS mutex `s_data_mutex`. Core 0 locks it for microseconds at 100 Hz; Core 1 takes a fast local snapshot at 1 Hz to prevent torn 32-bit float reads.	*"Dual-core chips don't need locks because tasks run on separate cores."* (Shared SRAM requires synchronization.)
21	What happens if a FreeRTOS task overflows its stack?	**A:** FreeRTOS stacks are fixed-size; we profiled watermarks and allocated 10 KB to `task_disaster_monitor` and 7 KB to `task_ppg_accelerator`, with `CONFIG_FREERTOS_CHECK_STACK_OVERFLOW` canary checking enabled.	*"FreeRTOS dynamically grows task stacks."* (It never grows stacks automatically.)
22	What is the bill-of-materials (BOM) cost in volume?	**A:** Approximately **$12–$15** for the core electronics in volume: ESP32-S3 (~$2.50), Renesas ForgeFPGA SLG47910 (~$0.50), plus sensors and OLED (~$9.00)—versus $50–$150 for a Zynq-7000 SoC alone.	*"The whole wearable costs $1."* (Quote realistic high-volume component pricing.)
23	What is the power consumption and expected battery runtime?	**A:** The ForgeFPGA draws **< 5 mW**. With the PMS5003 fan (~100 mA) and Wi-Fi SoftAP (~80 mA) duty-cycled, average system draw is **80–120 mA**, giving **10–14 hours** on a 1,200 mAh LiPo battery.	*"It runs for a month on a coin cell."* (Laser scattering fans and Wi-Fi require a LiPo pouch.)
24	Why does the UI show `settling n/240` for HRV during the first 3 minutes?	**A:** While preliminary RMSSD math starts at 30 beats, clinical time-domain HRV requires ~240 valid beat intervals (~3 minutes) for statistical stability. Meanwhile, HR, SpO_2, and mNEWS2 operate immediately because they do not depend on RMSSD.	*"The FPGA takes 3 minutes to lock onto the pulse."* (The FPGA locks in 80 ms; 240 beats is statistical windowing.)

#	Judge Question	20-Second Answer & Technical Follow-Up	Trap ("Don't Say")
25	How does the wearable handle cardiac arrhythmias?	**A:** The 250 ms refractory window tracks rates up to 240 BPM, and erratic beat intervals lower the Karlen SQI below 0.70, flagging the waveform and advising clinical ECG verification rather than publishing corrupted HRV.	*"Our TinyML model classifies atrial fibrillation."* (Optical PPG is screening, not a 12-lead ECG.)
26	How do you handle dark skin tones or calloused fingers?	**A:** Melanin and thick epidermis attenuate light. We configure a calibrated static LED current of **11.0 mA** (register `0x37`, safe ceiling 18.0 mA / `0x5A`) to maximize AC perfusion without thermal saturation, gated by Karlen SQI ≥ 0.70.	*"Skin pigmentation has zero effect on optical sensors."* (Acknowledge physics and explain the LED current + SQI design.)
27	Why is the stage demo powered over USB-C instead of a LiPo?	**A:** USB-C provides a rock-solid 5 V / 3.3 V rail for live Web-Serial oscilloscope telemetry and logic probing while eliminating lithium pouch puncture hazards during hackathon transport.	*"We forgot to design power regulation."* (Point to the 3.3 V LDO and LiPo header on the board.)
28	How do you distinguish a real fall from dropping the device on a desk?	**A:** A 5-state kinematic FSM requires free-fall ($< 0.60 \text{ g}$), high-impact deceleration ($> 2.50 \text{ g}$ and $\geq 80 \text{ °/s}$ within 800 ms), and continuous post-impact stillness ($\lvert A - 1 \text{ g} \rvert < 0.30 \text{ g}$ for ≥ 1.5 s), followed by a 10-second user cancel countdown.	*"Any hard bump triggers the siren."* (All three kinematic stages + the 10 s grace timer must complete.)
29	Can the FPGA hardware logic be updated in the field?	**A:** Yes. Because the SLG47910 is SRAM-based, the ESP32-S3 writes the 46,408-byte bitstream over SPI at every boot. An OTA firmware update updates both the C11 code and the FPGA hardware circuit simultaneously.	*"The FPGA is one-time programmable OTP."* (It is volatile SRAM configured by the MCU at boot.)
30	If a technical auditor attacked VALOR, what three weaknesses would you volunteer first?	**A:** (1) No core-equivalent temperature probe (fingertip skin estimate runs 1.23 °C low, so mNEWS2 temp is held at 0); (2) Synthetic RMSSD column in MIMIC-III v1.4 (58.18% sensitivity); (3) Fingertip PPG sensitivity to vigorous motion (mitigated by SQI < 0.70 gating).	*"There are no weaknesses in our system."* (Pre-empting the auditor with documented mitigations wins jury trust.)

Appendix A. Team Defense Roles & Chapter Ownership

Teammate	Primary Defense Role	Chapters Owned	Core Jury Topics to Master
Member 1	Live Hardware Demo, Sensors & Offline SoftAP	Ch 1, 2, 11, 13	5-act stage demo execution, I²C/UART sensor wiring, SpO_2 ratio-of-ratios, two-source skin temperature (+4.39 °C die offset & TMP117), 3-tier indoor GNSS fallback, and zero-install SoftAP `192.168.4.1`.

Teammate	Primary Defense Role	Chapters Owned	Core Jury Topics to Master
Member 2	FPGA Verilog RTL, SPI Interconnect & Timing	Ch 3, 4, 5, 6, 7, 12	Zynq-7000 vs. ForgeFPGA SLG47910, SPI Mode 0 framing & 5-byte `0xC3` burst, 2-FF CDC synchronizers, 8-tap boxcar filter (0-LUT `÷8`), 4-state Moore Peak FSM, 250 ms refractory math, CLB (75%) vs. LUT (45.98%), WNS +7.316 ns.
Member 3	FreeRTOS Firmware, INT8 TinyML & Clinical Rules	Ch 8, 9, 10, 14, 15	Core 0 vs. Core 1 task pinning & `s_data_mutex`, 619-byte INT8 MLP (sparsity audit & FPU dequantization), removal of PM2.5 neural net, MIMIC-III 58.18% sens / 98.19% spec, mNEWS2, 180-min cyclone regression, Fall & SOS FSMs.

Appendix B. Master Quick-Reference Card & 16-Question Self-Test

Subsystem	Key Numbers to Memorize Before Entering the Jury Room
SPI Bridge & CDC	4-wire SPI Mode 0 (`CPOL=0, CPHA=0`), 1 MHz runtime (8 µs/byte every 10 ms = 0.08% duty cycle; ≤ 5–10 MHz practical SCK limit under 50 MHz 2-FF CDC oversampling); 5-byte IBI burst starts with magic byte `0xC3`.
8-Tap Filter & Peak FSM	11-bit running sum (max $8 \times 255 = 2040$), `÷8` = wire shift `next_sum[10:3]` (0 LUTs), group delay 3.5 samples (35 ms), fill 80 ms. 4-state registered Moore FSM (2-bit binary): arm ≥ 120 + rising slope, confirm fall ≥ `peak_val >> 2` (25%), blank 12,500,000 cycles (250 ms = 240 BPM ceiling).
FPGA Synthesis & Timing	**ForgeFPGA SLG47910:** 515/1120 LUT5 (45.98%), 303/1856 FF (16.33%), 105/140 CLB (75.00% binding constraint), 0 DSP, 0 BRAM, 5/19 GPIO; WNS **+7.316 ns**, F_{max} **78.85 MHz**. **Zynq-7000:** 185 LUT, 266 FF, 0 DSP, 0 BRAM; OOC WNS **+5.603 ns** (69.45 MHz).
Dual-Core RTOS & AI	Core 0: `task_ppg_accelerator` (Pri 5, 7 KB, 100 Hz). Core 1: `task_disaster_monitor` (Pri 2, 10 KB, 1 Hz) + UART/IMU tasks (Pri 3, 4 KB) + Wi-Fi. TinyML: `6 → 24 → 16 → 3` (619 INT8 bytes vs. 2,476 B FP32; 391 zeros → effective `6 → 13 → 10 → 3`; 88.39% vs. 88.47%; 20–50 µs @ 160 MHz).
MIMIC-III & Safety Rules	16,387 records (98 patients): **94.11% acc, 98.19% spec, 58.18% sens**, 99.95% INT8 match. Forced CRITICAL if SpO_2 `< 88%`, `HR > 150`, `mNEWS2 ≥ 7`, or 180-min pressure slope `≤ −2.0 hPa/h` (≥ 0.5 hPa drop). SOS grace: 15 s vitals, 10 s fall, 5 s hold cancel, 30 s recovery.

Rapid Self-Test Checklist (With Verified Answers)

1. **Why does the FPGA exist when the ESP32-S3 is faster?** FreeRTOS has 5 ms tick jitter (10–50% HRV error); FPGA gives 20 ns hardware timestamps.
2. **Which edges sample and shift in SPI Mode 0?** Rising edge samples data (`CPHA=0`); falling edge shifts data; clock idles LOW (`CPOL=0`).
3. **What does MISO bit 7 mean, and why do bits [6:0] wrap?** Bit 7 is `beat_latched` (triggers 5-byte `0xC3` burst); 7 remaining bits wrap at 128 on diagnostic logs.
4. **Why pass SPI inputs through two flip-flops?** Prevents metastability when crossing from the asynchronous 1 MHz ESP32 domain into the 50 MHz FPGA domain.
5. **Why is the filter accumulator 11 bits wide?** $8 \times 255 = 2{,}040$, which requires $\lceil \log_2(2040) \rceil = 11$ bits to never overflow.
6. **Why is ÷8 free in hardware while ÷7 is expensive?** $\div 8$ is a 3-bit wire shift (0 LUTs); $\div 7$ requires combinational/sequential division logic.
7. **Name the 4 Peak-Detector FSM states and their transitions.** `ARMED (00)` → `RISING (01)` → `PEAK_FOUND (10)` → `REFRACTORY (11)`.
8. **How many 50 MHz clock cycles equal the 250 ms refractory window?** $50{,}000{,}000 \times 0.250 = 12{,}500{,}000$ cycles (caps HR at 240 BPM).
9. **What do WNS = +7.316 ns and TNS = 0 ps prove?** The slowest path finishes 7.316 ns before the 20 ns deadline (78.85 MHz F_{max}), with zero failing paths.
10. **Why is CLB usage 75% when LUT usage is only 45.98%?** Wide 32-bit counters span across 105/140 CLB tiles even when some LUT5s inside those tiles stay unused.
11. **Why does `s_data_mutex` guard the vitals struct?** Prevents torn 32-bit floating-point reads between the 100 Hz Core 0 writer and 1 Hz Core 1 reader.
12. **Verify the 619-byte parameter count of 6 → 24 → 16 → 3.** $L1: (6\times24)+24 = 168$; $L2: (24\times16)+16 = 400$; $L3: (16\times3)+3 = 51$; $168 + 400 + 51 = 619$ INT8 bytes.
13. **State the MIMIC-III result in one honest sentence.** 94.11% accuracy paired with **58.18% sensitivity / 98.19% specificity** across 16,387 ICU records (synthetic RMSSD caveat).
14. **Why was the PM2.5 neural network deleted?** It collapsed low smoke readings to zero in high humidity; replaced with US EPA / AirGradient polynomials.
15. **Why is the mNEWS2 temperature score held at 0?** A fingertip (30–34 °C) is not a core site, and the die estimator has a +4.39 °C LED offset and −1.23 °C reference gap.
16. **Trace a fall from motion to siren.** Free-fall ($<0.6g$) → Impact ($>2.5g$, $\geq 80°/s$ in 800 ms) → Stillness (1.5 s) → `SOS_ARMED` (10 s countdown) → `SOS_ACTIVE` (siren + OLED card).

Appendix C. Master Technical Glossary

Term	Definition in VALOR	Term	Definition in VALOR
ADC / ALU	Analog-to-Digital Converter (18-bit in MAX30102); Arithmetic Logic Unit in a CPU.	**Metastability**	Unstable flip-flop state hovering between 0 and 1 after a setup/hold violation on an async input.
AXI4-Lite	ARM AMBA memory-mapped 32-bit bus used on the Zynq-7000 baseline (`0x43C00000`).	**MIMIC-III v1.4**	MIT / Beth Israel Deaconess public critical-care database (16,387 records tested).
BME280	Bosch I²C sensor measuring ambient temperature, relative humidity, and barometric pressure.	**MLP / ReLU**	Multilayer Perceptron neural network; Rectified Linear Unit $f(x) = max(0, x)$.
BRAM / LUT	Block RAM (dedicated FPGA SRAM, 0/8 used); Look-Up Table (5-input logic cell on SLG47910).	**mNEWS2**	Modified National Early Warning Score 2 (0–3 points per vital; ≥ 7 forces CRITICAL).
CDC	Clock-Domain Crossing: safely passing signals between unrelated clocks via 2-FF synchronizers.	**MOSI / MISO**	Master-Out Slave-In and Master-In Slave-Out full-duplex SPI data lines.
CLB	Configurable Logic Block: physical slice containing 8 LUT5s and FFs on ForgeFPGA (105/140 used).	**MPU6050**	6-axis I²C MEMS inertial sensor (3-axis accelerometer + 3-axis gyroscope, polled at 40 Hz).
CPOL / CPHA	SPI Clock Polarity and Phase (Mode 0 = `0, 0` : idle low, sample rising, shift falling).	**NMEA-0183 / NVS**	ASCII GPS sentence format on UART2; Non-Volatile Storage flash partition on ESP32.
Dicrotic Notch	Secondary arterial pressure bounce after aortic valve closure; blanked by the 250 ms refractory state.	**PMS5003**	Plantower 90° laser particulate sensor on UART1 measuring PM1.0, PM2.5, and PM10.
DSP Slice / FPU	Hard multiplier block in FPGA (0 used); Floating-Point Unit in ESP32-S3 (runs dequantized MLP).	**PPG / SpO_2**	Photoplethysmography (optical blood-volume pulse); peripheral arterial oxygen saturation (%).
FF / RTL	Flip-Flop (1-bit clocked memory cell, 303 used); Register-Transfer Level hardware description.	**RMSSD (HRV)**	Root Mean Square of Successive Differences between IBIs (primary vagal HRV metric, in ms).
ForgeFPGA	Renesas SLG47910 ultra-low-power (< 5 mW), sub-dollar ($0.50) SRAM FPGA on ShrikeFi.	**SoftAP**	Software Access Point mode where ESP32-S3 hosts open Wi-Fi network `VALOR-0081` .
IBI	Inter-Beat Interval: time between consecutive systolic peaks (measured in 20 ns FPGA ticks).	**SQI (Karlen)**	Signal Quality Index (0.0–1.0) gating pulse morphology; HRV is withheld whenever $SQI < 0.70$.
INT8 / TinyML	8-bit signed integer weight quantization (`-128..127`) enabling sub-kilobyte MCU inference.	**WNS / TNS**	Worst Negative Slack (+7.316 ns) and Total Negative Slack (0 ps) in static timing analysis.